LEHAKOE THINANE (Leonard Kobuoe)

Systems Designer & Architect | AI Engineer

AI & Automation Systems · Backend & Integration Engineering · Cloud (Azure, AWS, OCI)
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Open to remote, hybrid and on-site client-embedded roles, including travel-based delivery.

PROFESSIONAL SUMMARY

Systems designer and architect who builds what he designs, with 10+ years delivering web and business systems and a current focus on AI-enabled platforms. Designed and built BiznizFlowPilot, a live multi-tenant ERP (94 PostgreSQL tables, 80 migrations, 1,380+ automated tests) covering inventory, purchasing, sales, invoicing, finance, HR/payroll and an AI assistant, and authored its HLD, LLD and technical dossier. By day, the embedded technical owner for consultants across 8+ subsidiary business units at a financial services group: working closely with the CTO on direction, running discovery, shaping the solution design, then building and deploying each system on .NET/C# with Azure and AWS, with roughly 75–80% of work centred on AI and automation. Designs AI that stays inside controlled boundaries (role-scoped tool access, audited data reads, provider abstraction and fallback, and human approval before changes reach live systems) and has applied this in a live ERP assistant, an approval-gated web agent and a .NET agent runtime. Works AI-natively, directing coding agents through structured instruction files and reviewing their output to production standards. Former agency team lead who chose to return to deep hands-on engineering.

CORE EXPERTISE

Architecture & Systems Design: Solution design, requirements-to-architecture translation, build-vs-integrate decisions, modular monolith design, multi-tenant data isolation, event-driven workflow processing, API and integration design, webhook architecture, technical design documentation (HLD/LLD)

AI & Intelligent Systems: LLM integration (Groq, OpenAI, Anthropic, Ollama local models), OpenAI function calling, tool-using AI assistants with role-scoped, audited data access, bounded agent execution (validated tool calls, step limits, verified outcomes), speech-to-text (Whisper) meeting summarisation, hybrid rule-based + LLM pipelines, confidence-scored recommendations, provider abstraction and fallback, human-in-the-loop and consent gating, n8n agent workflows

AI-Assisted Engineering: Agentic delivery with AI coding agents (Claude in VS Code): structured instruction files, investigate-then-plan approval gates, production-safety rules, and critical review of agent plans and generated code

Backend Engineering: Python, FastAPI, Pydantic, Celery, C#, .NET 9/10, ASP.NET Core, EF Core, Node.js, Express.js, PHP, TypeScript, REST APIs, webhooks

Automation & Integration: n8n, Azure Logic Apps, Office 365 / SharePoint automation, HubSpot, CRM/ERP integration, WordPress REST API

Data: PostgreSQL 17 (Supabase), SQL, relational data modelling, row locking and concurrency control, generated columns, partial indexes, atomic sequence allocation, query-performance tuning (N+1 elimination), SQLAlchemy, Alembic migrations, Redis, MySQL, MongoDB

Cloud & DevOps: Azure, AWS (EC2, S3, Lightsail), Oracle Cloud Infrastructure (networking, IAM, load balancers), Terraform, OCI CLI, Docker / Docker Compose, GitHub Actions CI, GHCR, Caddy, Nginx, Apache, Git / GitHub / Bitbucket

Security & Reliability: Authentication and session security, encryption, secrets management, audit logging, PII protection, JWT auth with bcrypt, role-based access, Sentry, health checks, pytest (80% coverage gate), xUnit, Ruff

Frontend: Next.js, React, Blazor, JavaScript (ES6+), HTML5, CSS3, Tailwind CSS, WordPress (custom themes and plugins, WooCommerce)

Earlier platforms: Adobe Experience Manager, Drupal, Shopify, Strapi, HubSpot CMS, SAP Commerce Cloud, Bynder DAM; PayFast, PayPal, Stripe

PROFESSIONAL EXPERIENCE

Web and AI Systems Developer | Tax Consulting South Africa **2024 – Present**

Embedded Solution Design | AI & Automation Engineering | .NET / Azure

- Act as the embedded technical owner for consultants across 8+ subsidiary business units: run discovery on each operational pressure point, translate it into a scoped solution design, then build and deploy it as the sole engineer on that project within a six-developer team.
- Work closely with the CTO, who sets technical direction: align each solution design and key architecture decision with the CTO before build, and draw on the CTO's guidance for prioritisation, trade-offs and review.
- Collaborate with fellow developers from time to time where projects share integrations, infrastructure or releases, including joint problem-solving and pre-release testing.
- Deliver 2–4 concurrent projects for 2–3 consultants at a time; roughly 75–80% involve AI and automation, built on .NET/C# and integrated with Azure and AWS infrastructure.
- Designed and deployed AI agents for the marketing function, covering social media management, SEO and web content production.
- Design integrations between newly built systems and existing subsidiary websites and marketing tools, and lead the migration of legacy subsidiary sites from WordPress to custom-coded applications.
- Build security controls into systems handling sensitive financial and client data (authentication, encryption, audit logging, secrets management and automated backups) to support regulatory compliance.
- Introduced a pre-release stress-testing step with a senior developer on every build, compensating for the absence of a dedicated QA function.

Web Developer & Automations | Baher Al-Damasy Group (Pty) Ltd **2022 – 2024**

Integration-First Automation | Multicloud Infrastructure | Vaal, South Africa & Egypt

- Owned automation and systems delivery for a multi-subsidiary group (hardware, hospitality, aviation school, import/export) with mirrored operations in South Africa and Egypt, working with a central Operations Manager to turn cross-business pressure points into planned, deployed solutions.
- Chose an integration-first approach, layering custom automations and AI workflows onto the group's existing HubSpot, CRM and ERP platforms instead of building replacements, to fit the centralised operating model and avoid duplicate systems.
- Provisioned and managed cloud infrastructure across OCI, AWS and Azure, including networking, IAM, monitoring, backups and security configuration.
- Codified infrastructure and environment provisioning with Terraform and OCI CLI, making deployments repeatable and consistent across environments.
- Coordinated with WordPress teams in South Africa and Egypt to integrate new automation into subsidiary websites.

Team Lead – Web & Development | EQ4 Digital Communication **2020 – 2022**

Technical Leadership | Solution Scoping | E-commerce Architecture | Rosebank, Johannesburg

- Led the Web Technologies and Software Development functions of a full-service agency, owning technical planning, task allocation and delivery across concurrent client and in-house projects.
- Architected a multi-tenant e-commerce marketplace for a government client in which independent vendors rent storefront space and manage their own catalogues through separate vendor dashboards.
- Scoped engagements directly from client briefs, choosing between WordPress and fully custom builds based on requirements, budget and long-term maintainability.
- Led design and implementation of custom plugins, themes and REST API integrations connecting custom systems to CMS-driven client sites.

Content Infrastructure & Digital Media Systems | Godlove Group (Pty) Ltd **2018 – 2020**

Automation | CRM Integration | Team Leadership | Fourways, Johannesburg

- Built automated customer-enquiry chatbots across web and social platforms for a Forex / Prop-Firm launch campaign, delivered against a fixed go-live date.
- Led a media department spanning web development, marketing and graphic design/video, aligning web systems with campaign launches.
- Built content-integration workflows connecting production output to CRM and marketing-automation platforms, with analytics tracking to optimise delivery.

Web and Content Developer | Hadithi Ya Africa **2015 – 2018**

Web Development | CMS & E-commerce Platforms | Rosebank, Johannesburg

- Built custom WordPress themes and supported Adobe Experience Manager implementations, turning design systems into responsive, cross-browser sites.
- Configured Drupal, Shopify, Strapi and HubSpot for marketing and headless commerce, and supported SAP Commerce Cloud and Bynder DAM integration on e-commerce projects.
- Maintained platform security and performance through updates, server-side tuning and proactive debugging.

SELECTED SYSTEMS & AI PROJECTS

BiznizFlowPilot | Multi-Tenant ERP SaaS (Founder & Engineer) app.mmnexus.co.za

Live business management platform for South African SMEs: CRM, multi-location inventory, purchasing, sales, invoicing with VAT, expenses and P&L, HR/payroll, documents, messaging, workflow automation and an AI assistant, with multi-country foundations. 94 PostgreSQL tables, 80 Alembic migrations, 1,380+ automated tests.

Architecture / Stack: Python, FastAPI, SQLAlchemy 2, Alembic, Supabase PostgreSQL 17, Celery + Redis, Next.js 16 / React 19 / TypeScript, Electron, Docker Compose, Caddy, Azure, GitHub Actions, Sentry, Cloudflare R2, Groq / Anthropic / Ollama, Whisper

- **Architecture:** modular monolith (one FastAPI application with domain modules layered API - service - repository over a shared PostgreSQL database), plus Celery workers and a separate meeting-agent service; authored the HLD, LLD and technical dossier.
- **Event-driven workflow engine:** events are persisted in the same transaction as the business change and claimed by workers with FOR UPDATE SKIP LOCKED; workflow runs snapshot their configuration; failures retry with capped backoff and stale work is recovered; stable action IDs act as idempotency keys for webhooks and email, with no false promise of exactly-once delivery.
- **Tenant isolation and financial integrity:** business-scoped repositories filter every query, and a CI guard fails the build on queries outside the repository layer; stock is reserved under row locks so concurrent orders can't oversell; gap-free invoice and payslip numbers from an atomic sequence; Decimal/NUMERIC money; P&L aggregated in SQL.
- **Performance:** with the API in South Africa and the database in Frankfurt (~350 ms per round trip), eliminated N+1 query patterns with eager loading, batched lookups and per-request caching, and added tests asserting query counts stay flat as data grows.
- **Controlled AI:** a tool-using in-app assistant whose tools run as the asking user, with that user's role and business scope, and log every data access; a provider-agnostic engine (Groq, Anthropic, Ollama) so models can be swapped or fall back; Whisper meeting summaries; and an AI meeting attendee built to join only after every participant consents.
- **Agentic delivery and production assurance:** built with AI coding agents steered by structured instruction files, approval gates and hard production-safety rules; directed an evidence-based production audit (backups, staging isolation, release integrity, idempotency, access) whose findings became fail-closed controls and release gates requiring named human sign-off for payroll payouts, Arabic/RTL and new-country launches.
- **Security and delivery:** JWT sessions in HttpOnly cookies, bcrypt, token revocation on password change, lockout and rate limiting, separate admin planes, encrypted per-tenant secrets; jurisdiction profiles (tax, currency, numbering, timezone, RTL Arabic) so new countries can be added without changing finance code; CI with Ruff, migrations on fresh PostgreSQL and an 80% coverage gate, images published to GHCR.

AI Growth Platform | Agentic Backend, .NET 10 (MVP) github.com/LehakoeThinane/ai-growth-platform

C#/.NET 10 agent runtime that runs growth-planning and incident-triage workflows through one shared, bounded executor.

Architecture / Stack: C#, ASP.NET Core, EF Core, PostgreSQL 17, OpenAI function calling, xUnit, GitHub Actions

- **Bounded execution:** one schema-validated tool call per step, with step limits, run deadlines and explicit failure states; write permission and record IDs come from the operator, never from the model.
- **Verified outcomes:** a run is marked complete only after the server re-checks what was saved; stale ticket updates are rejected by version checks; model advice is labelled as needing human review, and every step goes into an ordered audit trail.
- **Testability:** a deterministic demo mode tests every workflow without paid model calls, and PostgreSQL integration tests run in CI.

SEO-GEO Agent | Hybrid AI Agent / Multi-Tenant SaaS (MVP) github.com/LehakoeThinane

Crawls websites, detects SEO issues, generates fixes and applies approved changes to WordPress through a controlled Audit - Diagnose - Package - Approve - Apply pipeline.

Architecture / Stack: Python, FastAPI, background job queue, WordPress REST API, OpenAI / Groq / Ollama, Render

- **Hybrid design:** deterministic rules own issue detection, scoring and apply-safety decisions, while the LLM only improves recommendation wording, keeping outputs auditable and conservative.
- **Provider resilience:** LLM provider abstraction with automatic fallback (OpenAI - Ollama - Groq) and a heuristic engine as the final fallback, so recommendations continue when providers fail or are disabled.
- **Human-in-the-loop control:** every recommendation carries source and confidence and is classed safe-to-apply or review-required; changes are simulated before apply, and live writes are verified against the REST response and the rendered page.
- **SaaS foundations:** session authentication, tenants / sites / memberships, queued background jobs with stage-level progress, a health endpoint, and secure defaults that switch on automatically in hosted environments.

SELECTED ACHIEVEMENTS

- Improved order-processing efficiency by 25% with a React product configurator, built backend logic first and validated in production.
- Increased e-commerce conversions by 30% by pairing cart-abandonment automation with a redesigned checkout flow.
- Lifted average Google PageSpeed scores to 90+ across 10+ production client sites through backend and server-side performance tuning.